Drug design and development

Stages:

1) Identify target disease
2) Identify drug target
3) Establish testing procedures
4) Find a lead compound
5) Structure Activity Relationships (SAR)
6) Identify a pharmacophore
7) Drug design - optimising target interactions
8) Drug design - optimising pharmacokinetic properties
9) Toxicological and safety tests
10) Chemical development and production
11) Patenting and regulatory affairs
12) Clinical trials

- Have you ever had to take a medicine to treat an illness?

- Have you ever wondered how researchers determine if the medicines you take are safe or not?

Three main issues are involved in getting the drug to the market.

1. Drug has to be tested to ensure that it is safe and effective, and can be administered in a suitable fashion. This involves preclinical and clinical trials covering toxicity, drug metabolism, stability, formulation, and pharmacological tests.

2. Patenting and legal issues.

3. Drug has to be synthesized in ever-increasing quantities for testing and eventual manufacture. (chemical and process development).

Many of these issues have to be tackled in parallel.

The Drug Development Process

Step 1: Discovery and Development

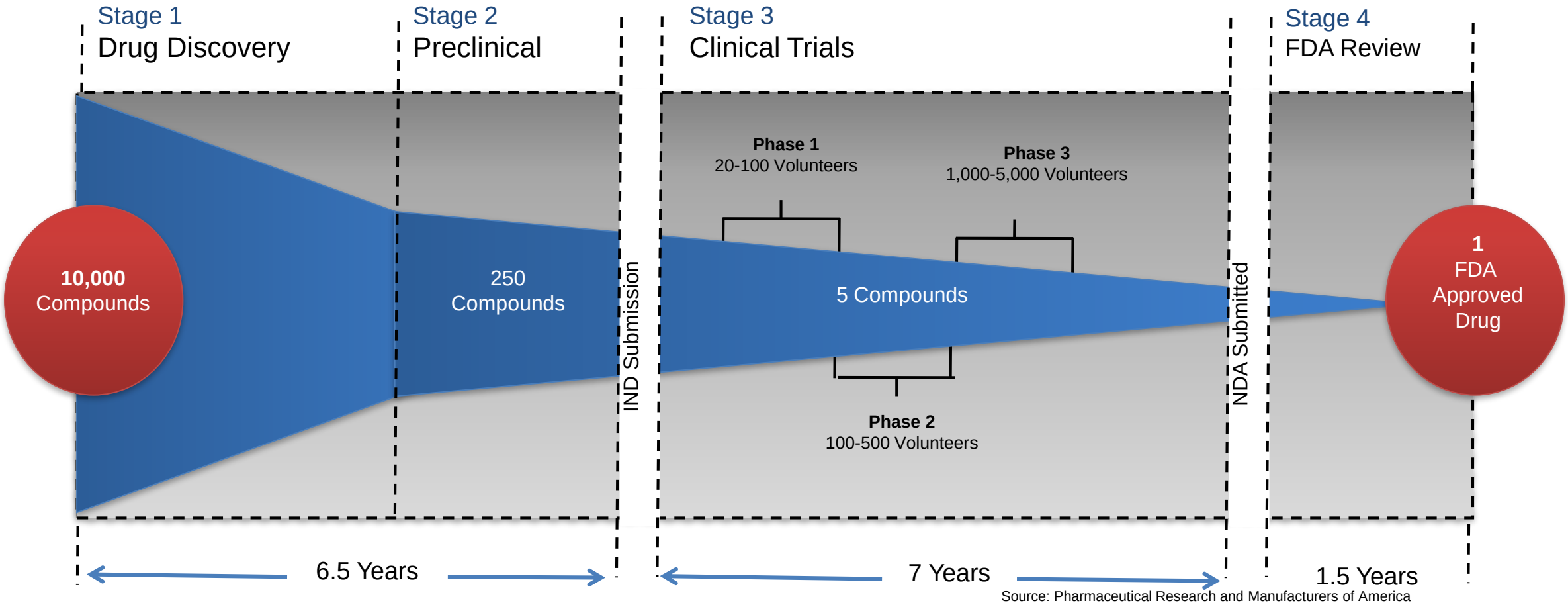
Step 2: Preclinical Research

Step 3: Clinical Research

Step 4: FDA Review

Step 5: FDA Post-Market Safety Monitoring

A Slow and Costly Process



Source: Pharmaceutical Research and Manufacturers of America

The average overall development cost of a new drug was recently estimated as $800 million or £444 million.

Pre-Clinical Studies

- In drug development, pre-clinical studies OR non-clinical studies , is a stage of research that begins before clinical trials (testing in humans), and during which important testing and drug safety data is collected.

- The main goals of pre-clinical studies are to determine a product's ultimate **safety profile** . Products may include new medical devices, drugs , gene therapy solutions , etc.

Preclinical studies involves:

1.Toxicity testing

2. Drug metabolism studies

3. Pharmacology, formulation, and stability tests

1.Toxicity testing

Test for carcinogenisity: *in vitro tests on* genetically engineered cell cultures and/or *in vivo testing* on transgenic mice to examine any eff ects on cell reproduction and to identify potential carcinogens.
Any signs of carcinogenicity would prevent the drug being taken any further studies.

Test for acute toxicity: administering sufficiently large doses *in vivo to produce a toxic* effect or death over a short period of time.

Further studies on acute toxicity then take place over a period of months, where the drug is administered to laboratory animals at a dose level expected to cause toxicity but not death.

Finally, long-term toxicology tests are carried out over a period of years at lower dose levels to test the drug for chronic toxic effects, carcinogenicity, special toxicology, mutagenicity, and reproduction abnormalities.

LD 50 : Value measuring drug toxicity which is the lethal dose required to kill 50% of a group of animals.

ED 50: dose required to produce the desired effect in 50% of test animals.

Therapeutic ratio or therapeutic index : The ratio of **LD 50** to **ED 50**

A therapeutic ratio of 10 indicates an LD 50 :ED 50 ratio of 10:1. This means that a 10-fold increase in the ED 50 dose would result in a 50% death rate.

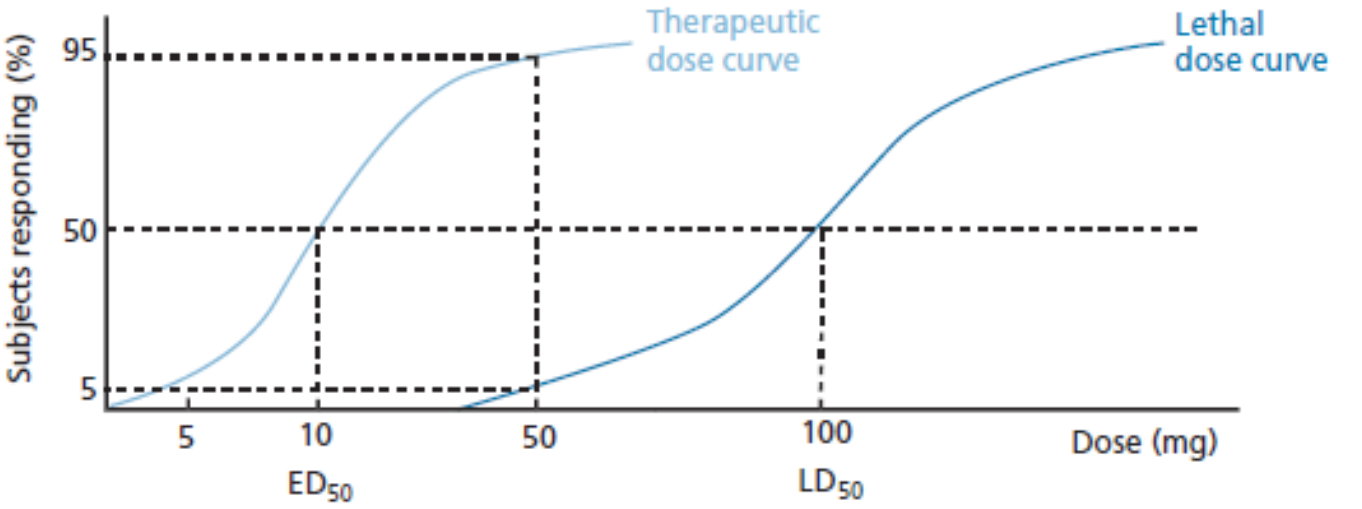


FIGURE 15.1 Comparison of therapeutic and lethal dose curves.

Ideally, the curves should not overlap on the x-axis, which means that the more gradual the two slopes, the riskier the drug will be

50-mg dose of the drug will act as a sedative for 95% of the test animals, but will be lethal for 5%. Such a drug would be unacceptable, even though it is eff ective in 95% of cases treated

2. Drug metabolism studies

- Find out what metabolites are formed from any new drug.
- The structure and stereochemistry of each metabolite.
- Biological activity of metabolite have to be tested .

Ideally, any metabolites that are formed should be inactive and quickly excreted

In vitro drug metabolism studies can also be carried out using perfused liver systems, liver microsomal fractions, or pure enzymes. Many of the individual cytochrome P450 enzymes that are so important in drug metabolism are now commercially available.

3. Pharmacology, formulation, and stability tests

Pharmacological tests involves the study of:

- Drug activity at targets other than the intended one.
- Drug’s mechanism of action.
- The a dose–response relationship.
- Drug’s duration of action.

Formulation studies involves:

- Developing a preparation of the drug which is both stable and acceptable to the patient.
- Testing drug compatibility with other tablet excepients.
- Characterization of a drug's physical, chemical, and mechanical properties in order to choose what other ingredients should be used in the preparation.
- Particle size, salt forms, crystal polymorphism, solvates, pH, and solubility, as all of these can influence bioavailability.

- Drug load: the ratio of the active drug to the total contents of the dose. A low drug load may cause homogeneity problems. A high drug load may pose flow problems or require large capsules if the compound has a low bulk density.

Stability studies involves:

- Effect of temperature, humidity, ultraviolet light, or visible light.
- Degradation products.
- Unwanted interactions between the preparation and the container.

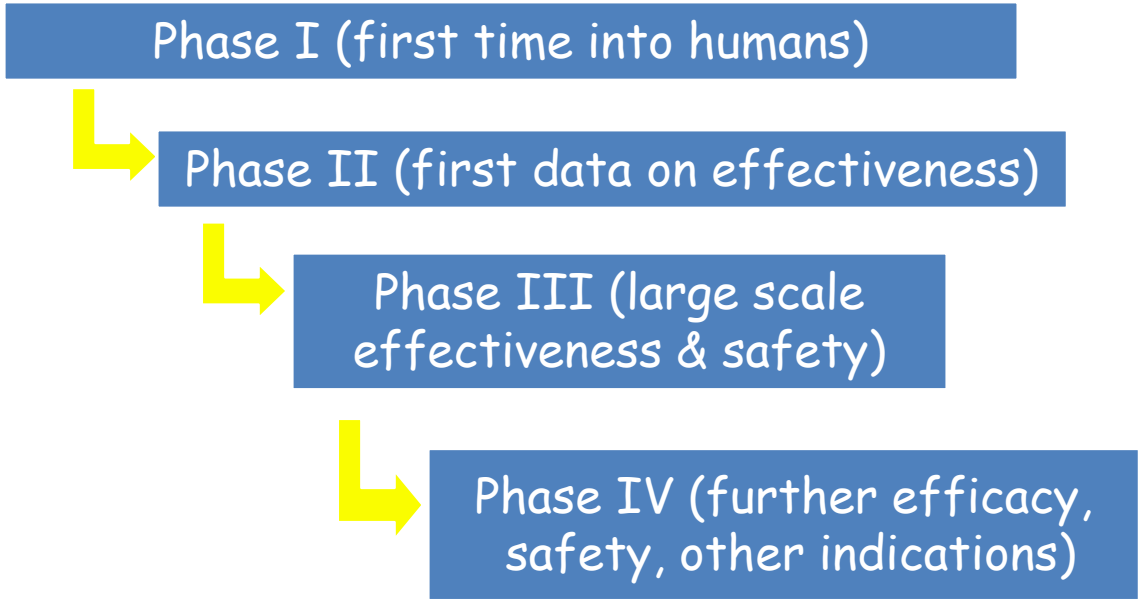
Clinical trials

- Clinical trials involve testing the drug on volunteers and patients, so the procedures involved must be ethical and beyond reproach.

- Involve hundreds to thousands of patients, and be extremely expensive.

- Can take 5–7 years to carry out,

Clinical studies



Phase I trials

- first introduction of IND (investigational new drug) in humans.
- Phase I studies take about a year and involve 100–200 volunteers.
- Usually only healthy adult volunteers (no patients).
- Provide a preliminary evaluation of the drug's safety, pharmacokinetics, and the dose levels that can be administered,
- Use dose-ranging (increasing dosages) to determine what side effects may occur.

- Check bioavailability (the fraction of administered drug that reaches the blood supply in a set period of time)

- Check bioequivalence (bioavailability remains the same be any alteration to the manufacturing, formulation, or storage processes).

Phase II trials

• Phase II studies generally last about two years and may start before phase I studies are complete.

•Test whether the drug has the therapeutic property claimed, pharmacokinetics and short-term safety of the drug, and to define the best dose regimen.

• Most phase II trials require 20–80 patients per dose group to demonstrate efficacy (200-500 volenterer).

• Initial trials (phase IIa) involve a limited number of patients to see if the drug has any therapeutic value at all, and to see if there are any obvious side effects

• Later studies (IIb) involve a larger numbers of patients. They are usually carried out as double-blind, placebo controlled studies.

double-blind, *placebo controlled studies*.: patients are split into two groups where one group receives the drug and the other group receives a placebo. In a double-blind study, neither the doctor nor the patient knows whether a placebo or drug is administered.

Phase III trials

- Take about three years on a much larger sample of patients (1000–5000 volunteers).
- These studies may begin before phase II studies are completed
- Phase IIIa studies
  Establish whether the drug is really effective or whether any beneficial effects are psychological.

  Allow further ‘tweaking’' of dose levels to achieve the optimum dose.
  Side effects not previously detected may be picked up with this larger sample of patients. If the drug succeeds in passing phase IIIa, it can be registered.

- Phase IIIb studies are carried out after registration, but before approval. They involve a comparison of the drug with those drugs that are already established in the field

Phase IV

- The drug is now placed on the market and can be prescribed, but it is still monitored for eff ectiveness and for any rare or unexpected side effects.

- This phase is a never-ending process as unexpected side effects may crop up many years after the introduction of the drug.

For example,
β-blocker **practolol** had to be withdrawn after several years of use because some patients suffered blindness and even death.

Rofecoxib (VIOXX) was used to treat rheumatoid arthritis for five years before a clinical trial carried out after its release showed that it was associated with increased risks of heart attack and stroke.

The drug was withdrawn voluntarily by Merck in 2004.

In the 5 years it had been on the market,

rofecoxib had been prescribed to 1.3 million patients in the USA and to 700,000 patients in 80 other countries.

Annual profits from the drug had reached $1.2 billion, which represented 18% of Merck's net income.

The loss of this income was so serious that the company's shareprice dropped 27% in value in a single day.

the company was faced with a lengthy litigation battle as thousands of patients sought compensation for alleged personal injuries resulting from the use of the drug

Ethical issues

- In phases I–III of clinical trials, the permission of the patient is mandatory.

- Ethical problems:

Unconscious patients, mentally ill patients and children.

Most licensed drugs have been licensed for adults, it means that around 40% of medicines given to children have never actually been tested on that age group.

Patenting and regulatory affairs

Patents

Exclusive rights to sell and manufacture its products for a reasonable period of time, and at a price which will not only recoup its costs, but that will generate sufficient profits for further research

In order to gain a patent,
1- submit or file the patent (what the new pharmaceutical is, what use it is intended for, and how it can be synthesized. This is no straightforward task).

2- The patent authorities decide whether the claims are novel and whether they satisfy the necessary requirements for that patent body.

It is important that a patent is filed as soon as possible.

- Patents in most countries run for 20 years after the date of filing. The protection period starts from the time of filing, not from when the drug comes onto the market.

Chiral switching : The issue of chiral switching relates mostly to racemic drugs that have been on the market for several years and are approaching the end of their patent life.
By switching to the pure enantiomer, companies can argue that it is a new invention and take out a new patent.

However, they have to prove that the pure enantiomer is an improvement on the original racemate and that they could not reasonably have been expected to know that when the racemate was originally patented.

Regulatory affairs:

a) The regulatory process

Before clinical trials can begin, the company has to submit the results of its scientific and preclinical studies to the relevant regulatory authority.

- Food and Drug Administration (FDA) in the USA
- Agency for the Evaluation of Medicinal Products (EMEA) in Europe

1- Investigational Exemption to a New Drug Application (IND)

(Confidential document submitted to the regulatory affair. The IND should contain information regarding the chemistry, manufacture, and quality control of the drug, as well as information on its pharmacology, pharmacokinetics, and toxicology).

2-The regulatory affair. assesses this information and then decides whether clinical trials can begin.

3- Any adverse results must be reported to the regulatory affair, who will discuss with the company whether the trials should be stopped.

4-If the clinical trials proceed smoothly, the company applies to the regulatory authority for marketing approval. This involves:

Submission of a **New Drug Application (NDA) to the FDA** in USA
Or Marketing Authorization Application (MAA) in Europe.

An NDA or MAA is typically 400–700 volumes in size, with each volume containing 400 pages! ! ! ! ! ! ! ! ! !

5- Once an NDA is approved, any modifications to a drug's manufacturing synthesis or analysis must be approved.

b) Fast-tracking and orphan drugs

Fast-tracking : drug reach the market as quickly as possible, smaller number of phase II and phase III clinical trials before the drug is put forward for approval.
IT is carried out for drugs that show promise for diseases where no current therapy exists and for drugs that show distinct advantages over existing ones in the treatment of life-threatening diseases as cancer.

Orphan drugs: are drugs that are effective against relatively rare medical problems. In the USA, an orphan drug is defined as one that is used for less than 200,000 people.

Because there is a smaller market for such drugs, pharmaceutical companies may be less likely to reap huge financial benefi ts and may decide not to develop and market an orphan drug.

c) Good laboratory, manufacturing, and clinical practice

Good Laboratory Practice (GLP) and Good Manufacturing Practice (GMP) are scientific codes of practice for a pharmaceutical company's laboratories and production plants.

They detail the scientific standards that are necessary and the company must prove to regulatory bodies that it is adhering to these standards.

Investigators involved in clinical research must demonstrate that they can carry out the work according to **Good Clinical Practice (GCP) regulations.**

d) Analysis of cost versus benefits

A medicine that is successfully licensed and reaches the market has one other hurdle to negotiate—a cost versus benefit analysis carried out by individual government authorities.

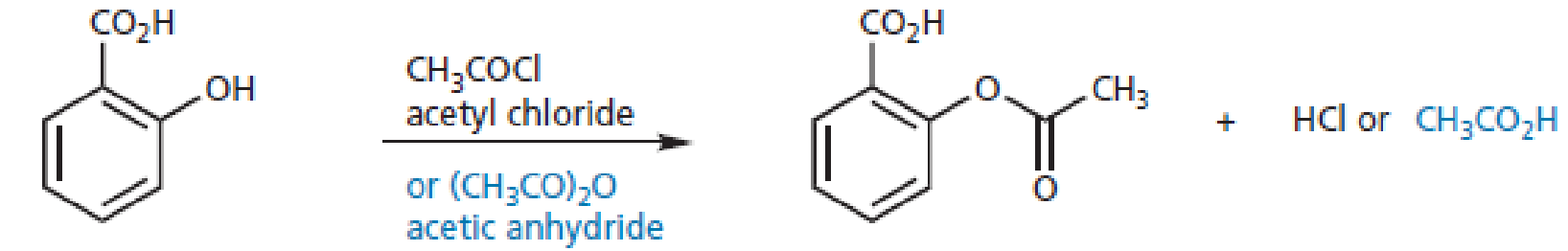
Chemical and process development

Chemical development

Once a compound goes forward for preclinical tests, it is necessary to start the development of a large-scale synthesis as soon as possible.

During chemical development, the conditions for each reaction in the synthetic route are studied closely and modified in order to get the best yields, purity, less cost and safety.

It is not necessary to scale up the synthetic route used by the research laboratories



Salicylic acid

Acetylsalicylic acid (aspirin)

Stages in chemical development

1- kilogram of drug is required for short-term toxicology and stability tests, analytical research, and pharmaceutical development.(original synthetic route)

2- 10 kg for long-term toxicology tests, as well as for formulation studies. Some of the material may also be used for phase I clinical trials.

3-100 kg is prepared for phase II and phase III clinical trials.

The **specifications of** the fi nal product are defined and determine the various analytical tests and purity standards required.

Process development

Process development aims to ensure that :

1-The number of reactions in the synthetic route is as small as possible

2- All the individual stages in the process are integrated with each other, such that the full synthesis runs smoothly and efficiently on a production scale.

3-Reduce the number of operations to a minimum.

4-Environmental and safety issues are extremely important.

5-Keeping costs low is a high priority and it is more economic to run the process such that a small number of large batches are produced rather than a large number of small batches.

6- Saftey

END OF LECTURE 3